

Question # 2 of 5 ( start time: 11:30:49 PM, 29 January 2023 )

T

$$T_{X \times Y}$$

denotes \_\_\_\_ on  $X \times Y$ .

Select the correct option

 Reload Math☐

product topology

☐

lower limit topology

☐

upper limit topology

☐

co finite topology

## Question # 4 of 5 ( Start time: 11:31:55 PM, 29 January 2023 )

Total M

Let  $X$  and  $Y$  be topological spaces. A map  $f : X \rightarrow Y$  is called an Open Map if.....

Select the correct option

 Reload Math Equations☐for every open set  $U \subseteq X$ , the image  $f(U) \subseteq Y$  is open.☐for every closed set  $U \subseteq X$ , the image  $f(U) \subseteq Y$  is closed.☐for every open set  $U \subseteq X$ , the image  $f(U) \subseteq Y$  is closed.☐

None of them

Question # 5 of 5 ( Start time: 11:32:23 PM, 29 January 2023 )

Total

Two spaces are called topologically equivalent if there exists a \_\_\_\_\_ between two topological spaces.

Select the correct option

☐

surjective map

☐

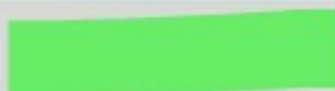
an injective map

☐

bijection

☐

homeomorphism



Question # 1 of 5 ( start time: 11:35:38 PM, 29 January 2023 )

Total Marks: 1

If  $(X, d)$  is a compact metric space, then\_\_\_\_\_

Select the correct option

 Reload Math Equations☐

every sequence is a Cauchy sequence.

☐

every sequence which has a convergent subsequence is a Cauchy sequence.

☐

every sequence has a subsequence which is a Cauchy sequence.

☐

every Cauchy sequence converges.

Question # 2 of 5 ( Start time: 11:36:30 PM, 29 January 2023 )

Total Marks: 1

A homeomorphic map between two topological spaces is always .....

select the correct option

|                                  |                      |
|----------------------------------|----------------------|
| <input type="radio"/>            | piecewise continuous |
| <input checked="" type="radio"/> | continuous           |
| <input type="radio"/>            | an identity map      |
| <input type="radio"/>            | discontinuous        |

## Question # 3 of 5 ( Start time: 11:37:16 PM, 29 January 2023 )

Total Marks: 1

Let  $X = \mathbb{R}$  (set of real numbers) be a usual metric space and  $N \subseteq \mathbb{R}$ , then which of the following is NOT true about  $N$  ?

Select the correct option

 Reload Math Equations

- |                       |                                                |
|-----------------------|------------------------------------------------|
| <input type="radio"/> | It is not a neighbourhood of any of its point. |
| <input type="radio"/> | None of them.                                  |
| <input type="radio"/> | It must be an open set.                        |
| <input type="radio"/> | Neither of its point is an interior point.     |

Question # 4 of 5 ( Start time: 11:39:47 PM, 29 January 2023 )

Total Marks: 1

Let  $X$  and  $Y$  be topological spaces. A map  $f : X \rightarrow Y$  is called a Closed Map if \_\_\_\_\_

Select the correct option

 Reload Math Equations

- |                       |                                                                               |
|-----------------------|-------------------------------------------------------------------------------|
| <input type="radio"/> | for every closed set $U \subseteq X$ , the image $f(U) \subseteq Y$ is closed |
| <input type="radio"/> | None of them                                                                  |
| <input type="radio"/> | for every open set $U \subseteq X$ , the image $f(U) \subseteq Y$ is closed.  |
| <input type="radio"/> | for every open set $U \subseteq X$ , the image $f(U) \subseteq Y$ is open.    |

Question # 5 of 5 ( Start time: 11:43:27 PM, 29 January 2023 )

Total Marks: 1

Which of the following statement is true?

Select the correct option

☐

Continuous map is always a closed map.

☐

none of these

☐

Continuous map may or may not be an open map

☐

Continuous map is always an open map.

A homoemorphic map between two topological spaces is always .....

Select the correct option

☐ piecewise continuous

☐ an identity map

☒ continuous



Pg9

☐ discontinuous

A map 'f' between two topological spaces is called .....  
iff 'f' is bijective, continuous and  $f^{-1}$  (f inverse) is continuous.

Select the correct option

☐ automorphism

☐ bijection

☐ isomorphism

☐ homeomorphism



Pg9

A map  $f$  from a topological space  $X$  to a topological space  $Y$  is \_\_\_\_ if and only if the image of each closed subset of  $X$  is closed in  $Y$ .

Select the correct option

|                                  |                  |
|----------------------------------|------------------|
| <input checked="" type="radio"/> | an open map      |
| <input type="radio"/>            | a closed map     |
| <input type="radio"/>            | an injective map |
| <input type="radio"/>            | a surjective map |

## Question # 4 of 5 ( Start time: 12:51:27 PM, 29 January 2023 )

Let  $X = \mathbb{R}$  (Set of real numbers) be a usual metric space and  $N \subseteq \mathbb{R}$ , then which of the following is true about  $N$  ?

Select the correct option

 Reload Ma

- |                       |                                                                   |
|-----------------------|-------------------------------------------------------------------|
| <input type="radio"/> | All of them.                                                      |
| <input type="radio"/> | It is not an open set.                                            |
| <input type="radio"/> | None of its point is an interior point.                           |
| <input type="radio"/> | It has disjoint intersection with the set of its interior points. |

## Homeomorphism:

A map  $f : X \rightarrow Y$  is called a **homeomorphism** between two topological spaces  $(X, \mathcal{T}_X)$  and  $(Y, \mathcal{T}_Y)$  iff

1.  $f$  is bijective.
2.  $f$  is continuous.
3.  $f^{-1} : Y \rightarrow X$  is continuous.

## morphism...

### Homeomorphic Spaces:

Two topological spaces  $(X, \mathcal{T}_X)$  and  $(Y, \mathcal{T}_Y)$  are called **homeomorphic** iff there exists a homeomorphism  $f : X \rightarrow Y$ . We also say such spaces **topologically equivalent** spaces.

$$X \approx Y$$

Restriction of an open map may or may not be an open map.